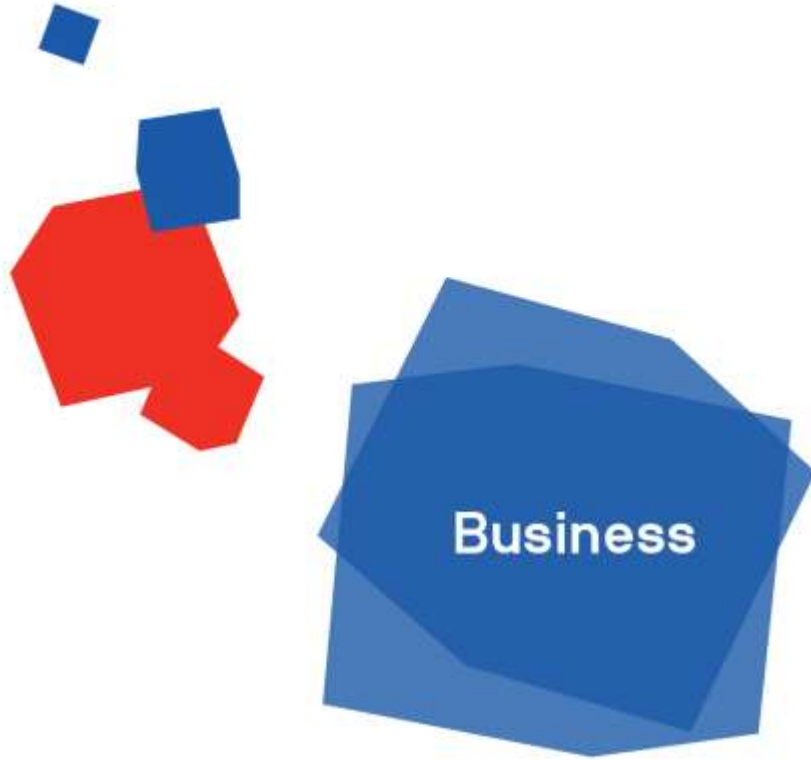


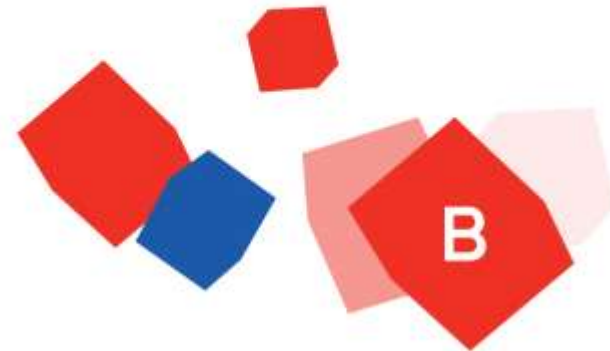


Fundamentals of Business Finance

Module 7



UNIVERSITY
OF TECHNOLOGY
SYDNEY



This week we will look at:

7.1 Overview of Equity - SLO2

7.2 Equity Financing Options - SLO2

7.3 Equity Valuation - SLO2

7.3 a Demonstration of share price calculation

Learning Objectives:

At the end of this module you will:

SLO2: apply time-value of money techniques to the valuation of financial instruments and financial securities

Characteristics of Equity

	Equity
Definition	• The difference between asset and debt. Equity refers to what belongs to shareholders. • Residual claim: Shareholders can only claim share of revenue after expenses (profit) if there is anything left. • In case of failure, shareholders paid last, if any funds remaining. Shareholders loss is limited to their investment. Creditors owed money cannot claim more from shareholders.
Payments	• Dividends are not committed. • Equity is not repayable
Tax deductibility	• Dividend are not a business expense. Paid after profit is taxed.
Voting power	• Can vote for Board Directors

Shareholder Rights

- **Dividend rights**
 - All shareholders receive a dividend per share.
 - $\text{Total Dividend} = \text{Dividend per Share} \times \text{number of shares held}$
- **Voting rights**
 - Vote for Board and/or significant changes in business
- **Asset rights**
 - Residual claim to assets after all debts have been paid
- **Participate in rights issue**
 - To avoid dilution of holdings



"Thank you gentlemen for voting me as chairman."

Dividends

$$\text{Equity} = \text{Contributed Capital} + \text{Retained Earnings} + \text{Reserves}$$

Dividends

- a **distribution of profit** from Retained Earnings
- **Paid at discretion** of company directors
- Paid from profits **after tax** (Retained Earnings)
- **not tax deductible**



Types of Equity



Ordinary Shares

- Standard equity
- Entitled to dividends if paid
- Voting rights
- Non-redeemable
- Residual claim
- Highest risk
- Highest return expected



Preference Shares

- Mix of Debt and Equity
- Fixed dividends (like interest) paid before ordinary dividends
- No voting rights
- Ranked higher than ordinary shares (prioritized above ordinary shareholders)
- Lower risk and return than ordinary shares

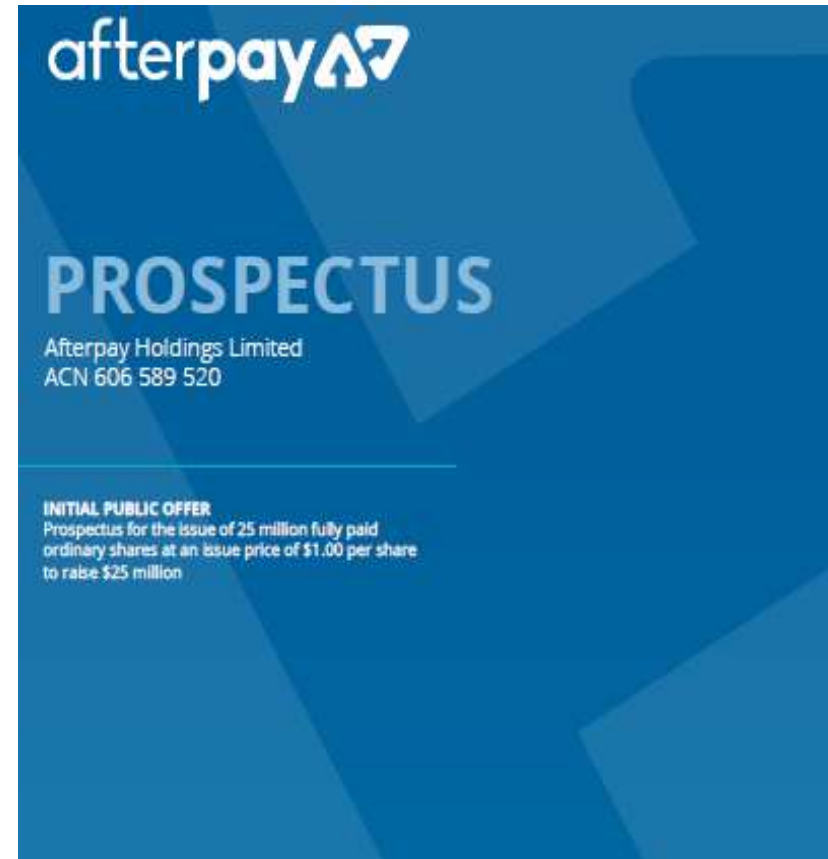
Questions

True or false ?

- Preference shares pay a fixed dividend rate to shareholders
 - **True**, preference shareholders will their dividend based on a fixed dividend payout rate.
- In case of business liquidation, Preference shareholders and Ordinary shareholders share the same priority in terms of claiming the residual asset after debtholders
 - **False**, Preference shareholders are prioritised before Ordinary shareholders in terms of claiming residual asset after Debtholders. This is one of the reasons why ordinary shareholders are taking more risks than preference shareholders.
- Both Preference shareholders and Ordinary shareholders have voting rights
 - **False**, since the preference shareholders receive the favour in receiving a fixed dividend rate, the voting rights are only reserved for ordinary shareholders who are exposed to more risks (dividend payments are not committed), compared to preference shareholders and debtholders
- Dividend payments are not tax deductible
 - **True**, dividend payments are paid after tax and they are not tax deductible. This is one of the reasons, some businesses prefer to borrow (as the interest expense is tax-deductible) over equity financing.
- Shareholders will receive their equity investment from the business when it's matured
 - **False**, unlike debt investment (loans, bills, bonds), equity investment is not to repaid to investors. Businesses might offer "share buyback" scheme where businesses are buying back shares from investors. However, this is totally up to the business management team's discretion, not an obligation

Initial public offering

- An initial public offering (IPO) is the initial issue of a firm's shares to the public
- A mechanism to **provide funds** for business owners
- IPOs are usually undertaken in conjunction with an **underwriter**, who advises on the pricing of the IPO (How much is the share price ?)
- If the IPO is not fully subscribed, the underwriter may be obligated to purchase the unsold shares.
- **Prospectus** required as offer to new investors



Stock Exchange Platforms

- Famous stock exchange platforms:



- In Australia: Australian Stock Exchange (ASX)



Advantages of being publicly listed

- Access to a **larger source of capital**, which facilitates future investment
- Improved **corporate image**
- Ease of attracting and retaining key personnel, align managers' goal with shareholders' (e.g. using share bonus as part of salary bonus)
- Market valuation

Disadvantages of being publicly listed

- Dilution of control (existing shareholders' ownership will be reduced)
- More disclosure and reporting requirements (e.g. the firm must appoint an auditor)
- Costly and lengthy process to raise funds

Issues with Initial Public Offerings

- Time-consuming
- Costly
- Uncertainty

To address the issues: Private placements

Private placement

- Private placements are off-market issues of new shares to individuals or groups
- May or may not be existing shareholders
- The investors are usually individuals or institutions who are regarded as friendly to the firm's directors



PLACEMENT AND NON-UNDERWRITTEN NON-RENOUNCEABLE RIGHTS ISSUE

The directors of Carpentaria Exploration Limited (ASX:CAP) (Carpentaria or the Company) are pleased to announce a placement to sophisticated and institutional investors to raise approximately A\$1,335,398 (Placement) and a non-underwritten non-renounceable entitlement issue to raise approximately A\$2,330,000 (before costs and expenses of the offer) (Rights Issue).

Advantages of private placement

- Quicker to implement and for smaller amounts
- They are much less costly (directly from the issuer to buyer)
- There is greater pricing certainty
- The new shares can be placed in friendly hands (choice in who you want to share the business with)

Disadvantages of private placements

- Dilute the interests of existing shareholders
- They are often unpopular with existing shareholders and can cause the share price to fall
- Cannot place over 15%

Why rights issues ?

Potential problem with raising funds through IPO

Dilution of control: The ownership of existing shareholders will reduce when new investors come in

Potential problem with raising funds through Private placement

There might be existing shareholders who are not provided with the opportunity participate in the equity raising scheme -> Unfair

To address the issue: right issues

Rights issue

A rights issue is an issue of rights to the **existing** shareholders of a firm, **based on their current holdings**.

With the right, **existing** shareholders are able to buy **ordinary shares** at a subscription price (offered by the issuing firm), which is normally cheaper than market price

May be underwritten

The terms of right issue is determined as follow:

$$\text{Terms of the right issue} = \frac{\text{Number of new shares}}{\text{Number of old outstanding shares}}$$

27 NOVEMBER 2015

TRANSURBAN RETAIL ENTITLEMENT OFFER

Details of a 1 for 18 pro rata accelerated renounceable entitlement offer of new Transurban stapled securities at an offer price of \$9.60 per security



RETAIL ENTITLEMENT OFFER
CLOSES AT 5.00PM (AEDT) ON
15 DECEMBER 2015

An example

Bran & Son wants to raise \$10 million in new equity. Suppose the subscription price is set at \$10 per share:

$$\text{Number of new shares} = \frac{\text{Funds to be raised}}{\text{Subscription price}} = \frac{10,000,000}{10} = 1,000,000$$

As there are 2 000 000 old shares outstanding, the terms of the rights issue is:

$$\text{Terms of the right issue} = \frac{\text{Number of new shares}}{\text{Number of old outstanding shares}} = \frac{1,000,000}{2,000,000} = \frac{1}{2}$$

Which means this is a 1-for-2 issue - that is, shareholders are entitled to one right for every two shares held currently

Features of rights issues

- Rights issues do not dilute the holdings of the firm's current shareholders
- Once a rights issue has been announced, shareholders have options to:
 - Exercise the rights to buy new shares at subscription price. As a result, their shareholding will not be diluted
 - Sell their rights to someone else
 - Do nothing – potentially lose out

Questions

The following true/false questions are to confirm your understanding on Equity financing options:

- In an IPO, shares can only be sold to a selected group of potential sellers
 - **False**, anyone in public can buy shares in an IPO event. However, companies can do equity issuing rounds pre-IPO. This is an opportunity for companies to sell their equity to targeted investors
- IPO is normally the cheapest option to raise funds through equity
 - **False**, to have a successful IPO, a company has to go through a lengthy process which is quite costly to them.
- Private placement suits best for business owners who want to have an idea of who are they sharing their business with
 - **True**, equity financing is where business owners sell part of their business (equity) in exchange for money. The new equityholders will now have access to the business so it's important to know who are you sharing the business with.

Questions

The following true/false questions are to confirm your understanding on equity raising through “Initial public offering”, “Private placement” and “Right issue”

- There is no limit on how much of equity can be sold through private placement
 - **False**, equity financing through private placement is capped at 15% of the total equity
- Right issue is designed to solve the dilution issue among existing shareholders
 - **True**, one of the biggest issues with IPO and/or Private placement is that the shareholding of existing shareholders might be diluted. This might cause dissatisfaction to existing shareholders which is not good. The right issue allows the existing shareholders the privilege to make a choice in whether or not they want to participate in this equity financing or not, before other potential shareholders.
- All shareholders will automatically receive new share through a right issue scheme
 - **False**, shareholders will receive a number of “rights” which is based on the number of existing shares they hold. The shareholders will then have to decide if they want to buy the new shares at “subscription price”, offered by the business. Shareholders, can totally sell the rights to someone else if they do not wish to participate in this fund raising round. However, at least they were provided with an opportunity

The PV of Common Stocks

- The value of any asset is the present value of its expected future cash flows.
- Stock ownership produces cash flows from:
 - Dividends
 - Capital Gains
- Valuation of Different Types of Stocks
 - Zero Growth
 - Constant Growth
 - Differential Growth

Case 1: Zero Growth

- ▶ Assume that dividends will remain at the same level forever

$$\text{Div}_1 = \text{Div}_2 = \text{Div}_3 = \dots$$

- Since future cash flows are constant, the value of a zero growth stock is the present value of a perpetuity:

$$P_0 = \frac{\text{Div}_1}{(1+R)^1} + \frac{\text{Div}_2}{(1+R)^2} + \frac{\text{Div}_3}{(1+R)^3} + \dots$$

$$P_0 = \frac{\text{Div}}{R}$$

Zero Growth Example

- Suppose Big Deal Company will pay an annual dividend of \$2.00 per common share that will never increase or decrease.
- The market rate of return is 8.5%.
- What is the maximum amount you should be willing pay for a common share of Big Deal Corporation?
- Formula for Zero Growth Model: $P = \text{Div} / R$
- Solution: $P = \$2.00 / .085$
 $P = \$23.53$

Case 2: Constant Growth

Assume that dividends will grow at a constant rate, g , forever, *i.e.*,

$$\text{Div}_1 = \text{Div}_0(1 + g)$$

$$\text{Div}_2 = \text{Div}_1(1 + g) = \text{Div}_0(1 + g)^2$$

$$\text{Div}_3 = \text{Div}_2(1 + g) = \text{Div}_0(1 + g)^3$$

$\vdots$

Since future cash flows grow at a constant rate forever, the value of a constant growth stock is the present value of a growing perpetuity:

$$P_0 = \frac{\text{Div}_1}{R - g}$$

Constant Growth Example

- Suppose Big D, Inc., just paid a dividend of \$.50. It is expected to increase its dividend by 2% per year. If the market requires a return of 15% on assets of this risk level, how much should the stock be selling for?
- $P_0 = .50(1+.02) / (.15 - .02) = \3.92

A Word About Dividends in the Constant Growth Model

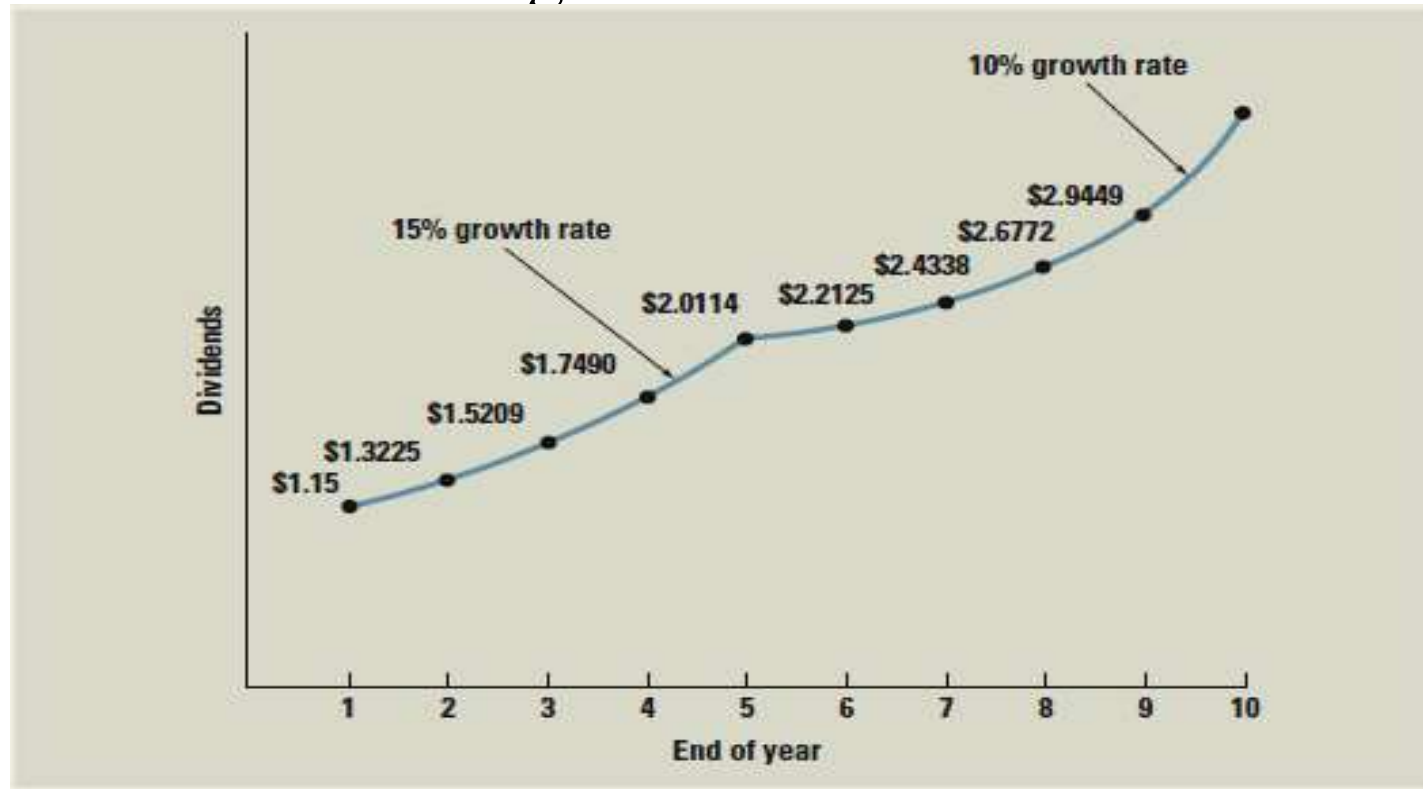
- It is critical to understand that in the constant growth model calculations are based on the **next** dividend
- If a situation only provides information on the last dividend it must be increased by the growth rate to arrive at the **next** dividend
- If a situation provides the value of the **next** dividend, then the data necessary for the calculation is known and need not be derived.
- An analyst must discriminate whether they have information about the **next** or last dividend and proceed with calculation accordingly

Case 3: Differential Growth

- ▶ Assume that dividends will grow at different rates in the foreseeable future and then will grow at a constant rate thereafter.
- ▶ To value a Differential Growth Stock, we need to:
 - Estimate future dividends in the foreseeable future.
 - Estimate the future stock price when the stock becomes a Constant Growth Stock (case 2).
 - Compute the total present value of the estimated future dividends and future stock price at the appropriate discount rate.

Graphic: Differential Dividend Growth

- This graph demonstrates the dividend profile for the example with differential growth



Case 3: Differential Growth

- Assume that dividends will grow at rate g_1 for N years and grow at rate g_2 thereafter.

$$\text{Div}_1 = \text{Div}_0(1 + g_1)$$

$$\text{Div}_2 = \text{Div}_1(1 + g_1) = \text{Div}_0(1 + g_1)^2$$

$$\vdots$$

$$\text{Div}_N = \text{Div}_{N-1}(1 + g_1) = \text{Div}_0(1 + g_1)^N$$

$$\text{Div}_{N+1} = \text{Div}_N(1 + g_2) = \text{Div}_0(1 + g_1)^N(1 + g_2)$$

$$\vdots$$

Case 3: Differential Growth

Dividends will grow at rate g_1 for N years and grow at rate g_2 thereafter

$$\begin{array}{c} \text{Div}_0(1+g_1) \quad \text{Div}_0(1+g_1)^2 \\ | \quad | \quad | \quad \dots \\ 0 \quad 1 \quad 2 \end{array}$$

$$\begin{array}{c} \text{Div}_0(1+g_1)^N \quad \text{Div}_N(1+g_2) \\ \dots \quad | \quad \text{---} \quad | \quad \dots \\ N \quad N+1 \end{array} = \text{Div}_0(1+g_1)^N(1+g_2)$$

Case 3: Differential Growth

We can value this as the sum of:

- a T -year annuity growing at rate g_1

$$P_A = \frac{C}{R - g_1} \left[1 - \frac{(1 + g_1)^T}{(1 + R)^T} \right]$$

- plus the discounted value of a perpetuity growing at rate g_2 that starts in year $T+1$

$$P_B = \frac{\left(\frac{\text{Div}_{T+1}}{R - g_2} \right)}{(1 + R)^T}$$

Case 3: Differential Growth

Consolidating gives:

$$P = \frac{C}{R - g_1} \left[1 - \frac{(1 + g_1)^T}{(1 + R)^T} \right] + \frac{\left(\frac{\text{Div}_{T+1}}{R - g_2} \right)}{(1 + R)^T}$$

Or, we can “cash flow” it out.

Example 1

A common stock just paid a dividend of \$2. The dividend is expected to grow at 8% for 3 years, then it will grow at 4% in perpetuity.

What is the stock worth? The discount rate is 12%.

With the Formula

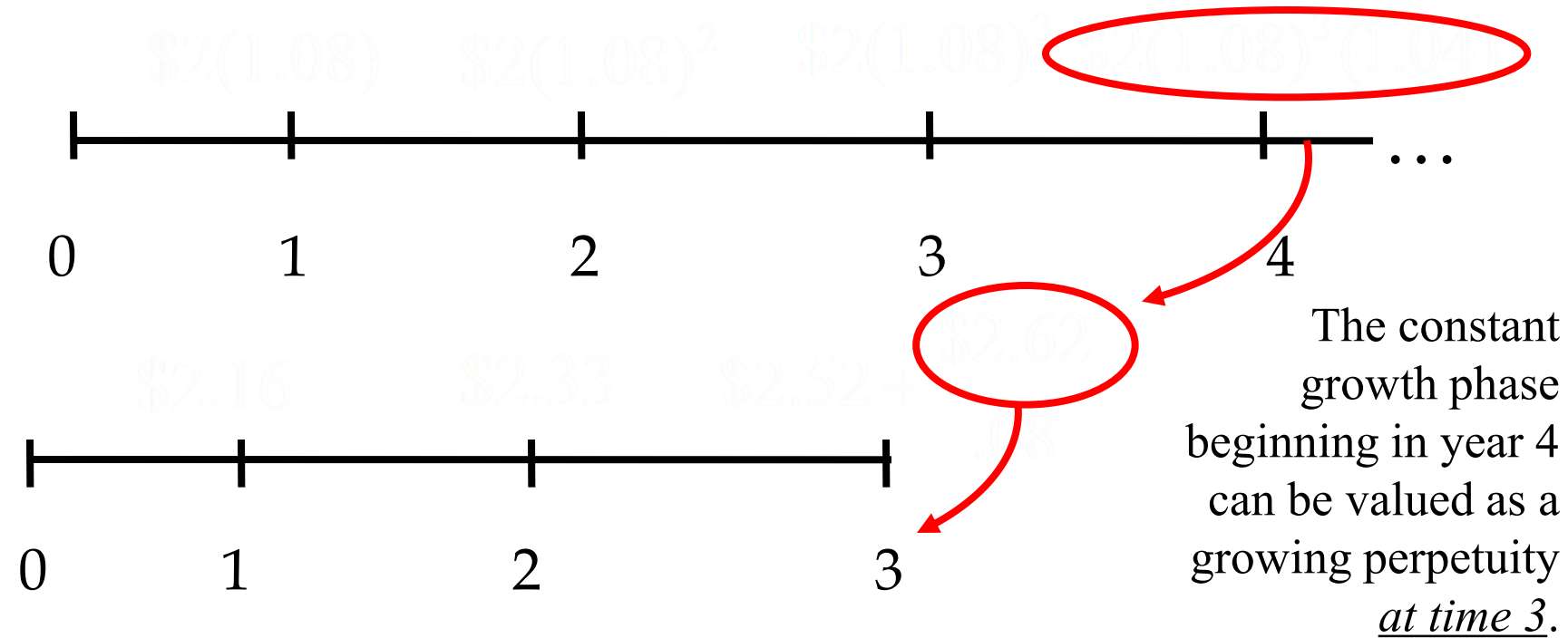
$$P = \frac{\$2 \times (1.08)}{.12 - .08} \left[1 - \frac{(1.08)^3}{(1.12)^3} \right] + \frac{\left(\frac{\$2(1.08)^3(1.04)}{.12 - .04} \right)}{(1.12)^3}$$

$$P = \$54 \times [1 - .8966] + \frac{(\$32.75)}{(1.12)^3}$$

$$P = \$5.58 + \$23.31$$

$$P = \$28.89$$

With Cash Flows



$$P_0 = \frac{\$2.16}{1.12} + \frac{\$2.33}{(1.12)^2} + \frac{\$2.52 + \$32.75}{(1.12)^3} = \$28.89$$

$$P_3 = \frac{\$2.62}{.08} = \$32.75$$

Example 2

- ❑ A company just paid a dividend of \$0.60
- ❑ Its required rate of return is 25%.
- ❑ The company expects its dividends to grow by 8% indefinitely
- What is the current value of the shares?

➤ Solution

$$P_0 = D_1 / (r - g)$$

$$P_0 = 0.60 (1 + 0.08) / (0.25 - 0.08)$$

$$P_0 = \$0.648 / 0.17$$

$$= \$3.81$$

Question 1

❑ What is the value of an ordinary share which just paid a dividend of \$2.00 and has an expected growth of 8% a year in perpetuity?

The required rate of return is 16%

a) \$27.00

b) \$12.50

c) \$25.00

d) \$13.50

The answer is a)

Comparables

- Comparables are used to value companies based primarily on multiples
- Common multiples include:
 - Price-to-Earnings
 - Enterprise Value Ratios

Price-Earnings Ratio

- The price-earnings ratio is calculated as the current stock price divided by annual EPS.
 - *The Wall Street Journal* uses last four quarters' earnings

$$\text{P/E ratio} = \frac{\text{Price per share}}{EPS}$$

Pricing with a comparable

- Suppose your company has EPS of \$1.69. The industry average P/E ratio is 26.94. What is the estimated price for your company?

$$P / E \times E = 26.94 \times 1.69 = \$45.53$$

Factors Impacting the P/E Ratio

- Generally, firms with growth opportunities command greater P/E than those with no such prospects
- A firm's R also impacts the P/E ratio. The P/E ratio and R are inversely related.
- A firm with conservative accounting principles will generally have a higher P/E ratio than one with aggressive policies

Enterprise Value Ratios

- The PE ratio focuses on equity, but what if we want the value of the firm?
- Use Enterprise Value:
 - $EV = \text{market value of equity} + \text{market value of debt} - \text{cash}$
- Like PE, we compare the value to a measure of earnings. From a firm level, this is EBITDA, or earnings before interest, taxes, depreciation, and amortization.
 - EBITDA represents a measure of total firm cash flow
- The Enterprise Value Ratio = $EV / EBITDA$

Enterprise Value Ratios

For example, imagine that Illinois Food Products Co. (IFPC) has equity worth \$800 million, debt worth \$300 million, and cash of \$100 million. The enterprise value here is \$1 billion ($= \$800 + 300 - 100$). Further imagine the firm has the following income statement:

ILLINOIS FOOD PRODUCTS CO. Income Statement (\$ In millions)	
Revenue	\$700.00
Cost of goods sold	<u>-500.00</u>
Earnings before interest, taxes, depreciation, and amortization (EBITDA)	\$200.00
Depreciation and amortization	- 100.00
Interest	<u>- 24.00</u>
Pretax income	76.00
Taxes (@ 30%)	<u>- 22.80</u>
Profit after taxes	\$ 53.20

The EV to EBITDA ratio is 5 ($= \$1 \text{ billion}/200 \text{ million}$). Note that all the items in the income statement below EBITDA are ignored when calculating this ratio.

As with PE ratios, it is generally assumed that similar firms have similar EV/EBITDA ratios. For example, imagine that the average EV/EBITDA ratio in an industry is 6. If QRT Corporation, a firm in the industry with EBITDA of \$50 million, is judged to be similar to the rest of the industry, its enterprise value might be estimated at \$300 million ($= 6 \times \50). Now imagine that QRT has \$75 million of debt and \$25 million of cash. Given our estimate of QRT's enterprise value, QRT's stock would be worth \$250 million ($= \$300 - 75 + 25$).

Valuing stocks using free cash flows

- We have discounted cash payouts to value a single share of stock and used the method of comparables.
- One can value stocks by discounting their cash flows using a “top down” approach.

Example 3:

As an example, consider Global Harmonic Control Systems (GHCS). Revenues, which are forecasted to be \$500 million in one year, are expected to grow at 10 percent per year for the two years after that, 8 percent per year for the next two years, and 6 percent per year after that. Expenses including depreciation are 60 percent of revenues. Net investment, including net working capital and capital spending less depreciation, is 10 percent of revenues. Because all costs are proportional to revenues, net cash flow (sometimes referred to as free cash flow) grows at the same rate as do revenues. GHCS is an all-equity firm with 12 million shares outstanding. A discount rate of 16 percent is appropriate for a firm of GHCS's risk.

YEAR (\$000,000)	1	2	3	4	5
Revenues	500.00	550.00	605.00	653.40	705.67
Expenses	300.00	330.00	363.00	392.04	423.40
Earnings before taxes	200.00	220.00	242.00	261.36	282.27
Taxes (40%)	80.00	88.00	96.80	104.54	112.91
Earnings after taxes	120.00	132.00	145.20	156.82	169.36
Net investment	50.00	55.00	60.50	65.34	70.57
Net cash flow	70.00	77.00	84.70	91.48	98.79

Example 3:

Since net cash flow grows at 6 percent per year after Year 5, net cash flow in Year 6 is forecasted to be \$104.72 ($= \98.79×1.06). Using the growing perpetuity formula, we can calculate the present value as of Year 5 of all future cash flows to be \$1,047.22 million [$= \$104.72 / (.16 - .06)$].

The present value as of today of that terminal value is:

$$\text{\$1,047.22} \times \frac{1}{(1.16)^5} = \text{\$498.59 million}$$

The present value of the net cash flows during the first five years is:

$$\frac{\text{\$70}}{1.16} + \frac{\text{\$77}}{(1.16)^2} + \frac{\text{\$84.7}}{(1.16)^3} + \frac{\text{\$91.48}}{(1.16)^4} + \frac{\text{\$98.79}}{(1.16)^5} = \text{\$269.39 million}$$

Adding in the terminal value, today's value of the firm is \$767.98 million ($= \$269.39 + 498.59$). Given the number of shares outstanding, the price per share is \$64.00 ($= \$767.98 / 12$).

The above calculation assumes a growing perpetuity after Year 5. However, we pointed out in the previous section that stocks are often valued by multiples. An investor might estimate the terminal value of GHCS via a multiple, rather than the growing perpetuity formula. For example, suppose that the price–earnings ratio for comparable firms in GHCS’s industry is 7.

Since earnings after tax in Year 5 are \$169.36. Using the PE multiple of 7, the value of the firm at Year 5 would be estimated as \$1,185.52 million ($= \169.36×7).

The firm’s value today is:

$$\frac{\$70}{1.16} + \frac{\$77}{(1.16)^2} + \frac{\$84.7}{(1.16)^3} + \frac{\$91.48}{(1.16)^4} + \frac{\$98.79}{(1.16)^5} + \frac{\$1,185.52}{(1.16)^5} = \$833.83$$

With 12 million shares outstanding, the price per share of GHCS would be \$69.49 ($= \$833.83/12$).

1. **Stock Values** The Herjavec Co. just paid a dividend of \$2.57 per share on its stock. The dividends are expected to grow at a constant rate of 4 percent per year indefinitely. If investors require a return of 11 percent on the stock, what is the current price? What will the price be in three years? In 15 years?

$$P_0 = D_0(1 + g)/(R - g)$$

$$P_0 = \$2.57(1.04)/(.11 - .04)$$

$$P_0 = \$38.18$$

$$P_3 = D_3(1 + g)/(R - g)$$

$$P_3 = D_0(1 + g)^4/(R - g)$$

$$P_3 = \$2.57(1.04)^4/(\cdot 11 - \cdot 04)$$

$$P_3 = \$42.95$$

$$P_{15} = D_{15}(1 + g)/(R - g)$$

$$P_{15} = D_0(1 + g)^{16}/(R - g)$$

$$P_{15} = \$2.57(1.04)^{16}/(\cdot 11 - \cdot 04)$$

$$P_{15} = \$68.77$$

10. Stock Valuation and Required Return Red, Inc., Yellow Corp., and Blue Company each will pay a dividend of \$2.86 next year. The growth rate in dividends for all three companies is 5 percent. The required return for each company's stock is 8 percent, 11 percent, and 14 percent, respectively. What is the stock price for each company? What do you conclude about the relationship between the required return and the stock price?

$$\text{Red stock price} = \$2.86 / (.08 - .05) = \$95.33$$

$$\text{Yellow stock price} = \$2.86 / (.11 - .05) = \$47.67$$

$$\text{Blue stock price} = \$2.86 / (.14 - .05) = \$31.78$$

As the required return increases, the stock price decreases. This is a function of the time value of money: A higher discount rate decreases the present value of cash flows. It is also important to note that relatively small changes in the required return can have a dramatic impact on the stock price.

29. Price—Earnings Ratio Consider Pacific Energy Company and U.S. Bluechips, Inc., both of which reported earnings of \$1.1 million. Without new projects, both firms will continue to generate earnings of \$1.1 million in perpetuity. Assume that all earnings are paid as dividends and that both firms require a return of 12 percent.

- a. What is the current PE for each company?
- b. Pacific Energy Company has a new project that will generate additional earnings of \$220,000 each year in perpetuity. Calculate the new PE ratio of the company.
- c. U.S. Bluechips has a new project that will increase earnings by \$440,000 each year in perpetuity. Calculate the new PE ratio of the company.

a. $P = D/R$

$$P = \$1,100,000/.12$$

$$P = \$9,166,666.67$$

The price-earnings ratio is the stock price divided by the current earnings, so the price-earnings ratio of each company with no growth is:

$$PE = \text{Price/Earnings}$$

$$PE = \$9,166,666.67/\$1,100,000$$

$$PE = 8.33 \text{ times}$$

- ▶ *b.* Since the earnings have increased, the price of the stock will increase. The new price of all the outstanding company stock is:



- ▶ $P = D/R$

- ▶ $P = (\$1,100,000 + 220,000)/.12$

- ▶ $P = \$11,000,000$



The price-earnings ratio is the stock price divided by the current earnings, so the price-earnings with the increased earnings is:



- ▶ $PE = \text{Price/Earnings}$

- ▶ $PE = \$11,000,000/\$1,100,000$

- ▶ $PE = 10 \text{ times}$

- c. Since the earnings have increased, the price of the stock will increase.

The new price of the all the outstanding company stock is:

$$P = D/R$$

$$P = (\$1,100,000 + 440,000)/.12$$

$$P = \$12,833,333.33$$

The price-earnings ratio is the stock price divided by the current earnings, so the price-earnings with the increased earnings is:

$$PE = \text{Price/Earnings}$$

$$PE = \$12,833,333.33/\$1,100,000$$

$$PE = 11.67 \text{ times}$$

- 31. Stock Valuation and EV** Lotharan Corp. has yearly sales of \$31.5 million and costs of \$17.3 million. The company's balance sheet shows debt of \$59 million and cash of \$21 million. There are 975,000 shares outstanding, and the industry EV/EBITDA multiple is 7.5. What is the company's enterprise value? What is the stock price per share?

$$\text{EBITDA} = \text{Sales} - \text{Costs}$$

$$\text{EBITDA} = \$31,500,000 - 17,300,000$$

$$\text{EBITDA} = \$14,200,000$$

$$\text{Enterprise value} = \$14,200,000(7.5)$$

$$\text{Enterprise value} = \$106,500,000$$

$$\text{Equity value} = \text{Enterprise value} - \text{Debt} + \text{Cash}$$

$$\text{Equity value} = \$106,500,000 - 59,000,000 + 21,000,000$$

$$\text{Equity value} = \$68,500,000$$

So, the price per share is:

$$\text{Stock price} = \$68,500,000 / 975,000$$

$$\text{Stock price} = \$70.26$$

- 32. Stock Valuation and Cash Flows** Maslyn Manufacturing has projected sales of \$153 million next year. Costs are expected to be \$82 million, and net investment is expected to be \$15 million. Each of these values is expected to grow at 14 percent the following year, with the growth rate declining by 2 percent per year until the growth rate reaches 6 percent, where it is expected to remain indefinitely. There are 6.2 million shares of stock outstanding and investors require a return of 12 percent return on the company's stock. The corporate tax rate is 40 percent.
- a. What is your estimate of the current stock price?
 - b. Suppose instead that you estimate the terminal value of the company using a PE multiple. The industry PE multiple is 12. What is your new estimate of the company's stock price?

		Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
a.	Sales	\$153,000,000	\$174,420,000	\$195,350,400	\$214,885,440	\$232,076,275	\$246,000,852
	Costs	82,000,000	93,480,000	104,697,600	115,167,360	124,380,749	131,843,594
	EBT	\$71,000,000	\$80,940,000	\$90,652,800	\$99,718,080	\$107,695,526	\$114,157,258
	Taxes	28,400,000	32,376,000	36,261,120	39,887,232	43,078,211	45,662,903
	Net income	\$42,600,000	\$48,564,000	\$54,391,680	\$59,830,848	\$64,617,316	\$68,494,355
	Investment	15,000,000	17,100,000	19,152,000	21,067,200	22,752,576	24,117,731
	Cash flow	\$27,600,000	\$31,464,000	\$35,239,680	\$38,763,648	\$41,864,740	\$44,376,624

To find the terminal value of the company in Year 6, we can discount the Year 7 cash flows as a growing perpetuity, which will be:

$$\text{Terminal value} = \$44,376,624(1 + .06)/(.12 - .06)$$

$$\text{Terminal value} = \$783,987,028$$

So, the value of the company today is:

$$\begin{aligned} \text{Company value today} = & \$27,600,000/1.12 + \$31,464,000/1.12^2 + \$35,239,680/1.12^3 \\ & + \$38,763,648/1.12^4 + \$41,864,740/1.12^5 + (\$44,376,624 + 783,987,028)/1.12^6 \end{aligned}$$

$$\text{Company value today} = \$542,873,656$$

Dividing the company value by the shares outstanding to get the share price, we get:

$$\text{Share price} = \$542,873,656/6,200,000 \quad \text{Share price} = \$87.56$$

- ▶ *b.* In this case, we are going to use the PE multiple to find the terminal value. All of the cash flows from part *a* will remain the same. So, the terminal value in Year 6 is:

$$\text{Terminal value} = 12(\$68,494,355)$$

$$\text{Terminal value} = \$821,932,257$$

Under this assumption for the terminal value, the value of the company today is:

$$\text{Company value today} = \$27,600,000/1.12 + \$31,464,000/1.12^2 + \$35,239,680/1.12^3 + \$38,763,648/1.12^4 + 41,864,740/1.12^5 + (\$44,376,624 + 821,932,257)/1.12^6$$

$$\text{Company value today} = \$562,097,890$$

Dividing the company value by the shares outstanding to get the share price we get:

$$\text{Share price} = \$562,097,890/6,200,000$$

$$\text{Share price} = \$90.66$$

34. Stock Valuation Most corporations pay quarterly dividends on their common stock rather than annual dividends. Barring any unusual circumstances during the year, the board raises, lowers, or maintains the current dividend once a year and then pays this dividend out in equal quarterly installments to its shareholders.

- a. Suppose a company currently pays an annual dividend of \$2.96 on its common stock in a single annual installment, and management plans on raising this dividend by 4 percent per year indefinitely. If the required return on this stock is 11 percent, what is the current share price?
- b. Now suppose that the company in (a) actually pays its annual dividend in equal quarterly installments; thus, this company has just paid a dividend per share of \$.74, as it has for the previous three quarters. What is your value for the current share price now? (*Hint:* Find the equivalent annual end-of-year dividend for each year.) Comment on whether or not you think this model of stock valuation is appropriate.

- ▶ *a.* Using the constant growth model, the price of the stock paying annual dividends will be:

$$P_0 = D_0(1 + g)/(R - g)$$

$$P_0 = \$2.96(1.04)/(.11 - .04)$$

$$P_0 = \$43.98$$

b. If the company pays quarterly dividends instead of annual dividends, the quarterly dividend will be one-fourth of the annual dividend, or:

$$\text{Quarterly dividend} = \$2.96(1.04)/4$$

$$\text{Quarterly dividend} = \$0.77$$

To find the equivalent annual dividend, we must assume that the quarterly dividends are reinvested at the required return. We can then use this interest rate to find the equivalent annual dividend. In other words, when we receive the quarterly dividend, we reinvest it at the required return on the stock. So, the effective quarterly rate is:

$$\text{Effective quarterly rate} = 1.11^{.25} - 1$$

$$\text{Effective quarterly rate} = .0264, \text{ or } 2.64\%$$

The effective annual dividend will be the FVA of the quarterly dividend payments at the effective quarterly required return. In this case, the effective annual dividend will be:

$$\text{Effective } D_1 = \$0.77(\text{FVIFA}_{2.64\%, 4})$$

$$\text{Effective } D_1 = \$3.20$$

Now, we can use the constant growth model to find the current stock price as:

$$P_0 = \$3.20 / (.11 - .04)$$

$$P_0 = \$45.75$$